Multi-Sensor Fusion for Autonomous Driving Perception: A Comprehensive Survey

Abstract

Autonomous driving perception systems rely heavily on the integration of heterogeneous sensor modalities to achieve robustness, accuracy, and safety in complex dynamic environments. While single-sensor systems face inherent limitations—such as LiDAR sparsity at long ranges, camera vulnerability to adverse lighting, and radar ambiguity in spatial resolution—multi-sensor fusion (MSF) offers a pathway to complementary redundancy. This survey provides a comprehensive review of the state-of-the-art in multi-sensor fusion for autonomous driving, synthesizing insights from 300 recent academic studies. We present a detailed taxonomy of fusion architectures, ranging from early data-level integration to late decision-level aggregation, with a specific focus on emerging Bird's-Eye-View (BEV) representations, transformer-based attention mechanisms, and uncertainty-aware fusion strategies. Furthermore, we examine the critical roles of cooperative perception (V2X), novel sensor modalities including 4D imaging radar and event cameras, and the growing importance of robustness against sensor corruption, adversarial attacks, and temporal misalignment. By analyzing methodological innovations, dataset advancements, and persistent challenges, this paper outlines the current landscape and future directions for deploying reliable multi-modal perception systems in real-world autonomous vehicles.

1. Introduction

The realization of fully autonomous driving (Level 4 and Level 5) necessitates perception systems capable of operating reliably under diverse and often adverse conditions. No single sensor modality possesses the comprehensive capabilities required for such tasks. LiDAR provides precise geometric depth but suffers from performance degradation in heavy precipitation and lacks semantic texture [1], [2]. Cameras offer rich semantic information and high resolution but are susceptible to illumination changes, glare, and darkness [3], [4]. Radar sensors provide robust velocity measurements and all-weather operation but generate sparse, noisy point clouds with limited elevation resolution [5], [6]. Consequently, Multi-Sensor Fusion (MSF) has emerged as the cornerstone of modern autonomous perception, aiming to leverage the complementary strengths of these modalities while mitigating their individual weaknesses [7], [8].

The evolution of MSF has progressed from simple heuristic combinations to sophisticated deep learning architectures. Early approaches often relied on late fusion, where independent detections were merged using Kalman filters or non-maximum suppression [9], [10]. However, the advent of deep neural networks enabled feature-level and data-level fusion, allowing models to learn cross-modal correlations directly from raw or intermediate representations [11], [12]. Recent years have witnessed a paradigm shift towards unified Bird's-Eye-View (BEV) representations, where features from cameras, LiDAR, and radar are projected into a common spatial reference frame, facilitating seamless integration and multi-task learning [13], [14]. Simultaneously, the introduction of transformer architectures has revolutionized fusion by enabling soft, attention-based association of features across modalities without requiring perfect spatial alignment [15], [16].

Despite significant progress, several critical challenges remain. Sensor fusion systems must handle temporal misalignment caused by asynchronous sampling rates [17], [18], extrinsic calibration errors [19], [20], and sensor failures or corruptions [21], [22]. Furthermore, the integration of novel sensors such as 4D imaging radar [23], [24] and event cameras [25], [26] introduces new opportunities and complexities. The rise of cooperative perception (V2X) extends the perceptual horizon beyond line-of-sight but demands robust communication and alignment protocols [27], [28]. Security concerns, including adversarial attacks that exploit cross-modal inconsistencies, also pose significant threats to system integrity [29], [30].

This survey synthesizes findings from 300 relevant academic papers to provide a holistic view of the MSF landscape. We categorize methodologies based on fusion stages, representation spaces, and architectural paradigms. We extensively cover the transition from traditional CNN-based pipelines to transformer-driven frameworks, the emergence of uncertainty quantification for robust fusion, and the critical role of large-scale multimodal datasets. By comparing methodological families and identifying open problems, this paper aims to guide future research towards more resilient, efficient, and scalable autonomous driving perception systems.

2. Method

The methodology of multi-sensor fusion can be systematically categorized based on the stage at which information from different modalities is combined, the representation space used for integration, and the underlying architectural mechanisms. This section provides a detailed taxonomy and analysis of these methodological families, supported by extensive evidence from recent literature.

2.1. Fusion Stages and Architectural Paradigms

The timing of fusion within the perception pipeline fundamentally dictates the trade-off between information preservation and computational complexity. Traditional taxonomies divide fusion into early (data-level), intermediate (feature-level), and late (decision-level) strategies [7], [31], [12].

2.1.1. Early and Data-Level Fusion Early fusion involves combining raw sensor data or low-level features before significant processing. This approach maximizes information retention but requires precise spatiotemporal alignment and homogeneous data structures. * **Raw Data Concatenation:** Methods like RVF-Net project camera RGB and radar velocity features directly onto LiDAR points prior to voxelization, enabling the network to learn interdependencies between sparse radar data and dense LiDAR features from the earliest layers [32]. Similarly, some frameworks concatenate RGB and depth images (derived from LiDAR) into a single input tensor for standard CNNs, treating depth as an additional channel to enable joint detection in a single pass [33], [34]. * **Challenges:** While effective, early fusion is highly sensitive to calibration errors. Misalignment at the input level can propagate through the network, degrading performance significantly [19]. Furthermore, fusing heterogeneous data types (e.g., dense images vs. sparse point clouds) often requires complex projection operations that may introduce quantization artifacts [11].

2.1.2. Intermediate and Feature-Level Fusion Intermediate fusion, currently the dominant paradigm, aggregates features extracted by modality-specific backbones. This balances the preservation of modality-specific characteristics with the ability to learn cross-modal interactions. * **BEV-Centric Fusion:** A major trend is the projection of all modalities into a unified Bird's-Eye-View (BEV) space. BEVFusion exemplifies this by disentangling camera and LiDAR encoders, projecting their features into a shared BEV grid, and fusing them via simple concatenation or attention mechanisms [13], [35]. This approach preserves the geometric structure of LiDAR and the semantic density of cameras without the information loss associated with direct perspective-to-BEV projection. * **Cross-Modal Attention:** Transformer-based architectures have introduced soft association mechanisms. FUTR3D employs a Modality-Agnostic Feature Sampler (MAFS) that uses object queries to sample features from any available modality, allowing a single architecture to handle arbitrary sensor combinations [16]. Similarly, methods like DeepFusion utilize lidar-guided projection to transform image features into BEV, followed by explicit feature alignment modules to harmonize spatial and semantic discrepancies [36]. * **Hierarchical Fusion:** Some approaches adopt a cascaded strategy. For instance, Multi-Modality Cascaded Fusion performs intra-frame association on high-confidence detections followed by global association on residuals, preserving interpretability while leveraging complementary strengths [37].

2.1.3. Late and Decision-Level Fusion Late fusion combines independent detection results or tracks from each modality. This approach offers modularity and robustness to single-sensor failures but may miss subtle cross-modal cues. * **Track-to-Track Association:** Systems like VALISENS implement hierarchical late fusion, aggregating intra-entity sensor data via track-to-track Kalman filtering and exchanging Collective Perception Messages (CPM) over V2X links [38]. RMMDet utilizes result-level fusion, matching 3D LiDAR pillars with 2D camera boxes via distance-sorted soft association, ensuring robustness against single-sensor failure [39]. * **Uncertainty-Aware Meta-Classification:** YOdar processes camera and radar data independently and fuses them using a gradient boosting classifier on geometric and statistical meta-features, leveraging uncertainty metrics to weight sensor contributions [40]. While robust, late fusion often struggles with conflicting detections and lacks the end-to-end optimization benefits of deeper fusion strategies [41].

2.2. Representation Spaces and View Transformations

The choice of representation space is critical for aligning heterogeneous sensor data. The field has evolved from perspective-view projections to unified 3D and BEV representations.

2.2.1. Bird's-Eye-View (BEV) Representations BEV has become the de facto standard for multi-sensor fusion due to its ability to provide a scale-invariant, occlusion-aware representation of the scene. * **View Transformation Techniques:** Converting perspective camera images to BEV is a core challenge. Lift-Splat-Shoot (LSS) predicts depth distributions to lift 2D features into 3D space before splatting them onto a BEV grid [14], [42]. Advanced methods like BEVFusion4D utilize LiDAR BEV as a spatial prior to guide deformable cross-attention for camera feature projection, ensuring semantic-spatial alignment [43]. * **Unified BEV Encoders:** UniBEV employs uniform deformable attention-based BEV encoders for both camera and LiDAR, generating spatially aligned feature maps that eliminate the need for distinct branch designs and improve robustness to missing modalities [44]. BroadBEV further refines this by scattering LiDAR BEV density to camera depth distributions, guaranteeing geometric synchronization [45].

2.2.2. Voxel and Point-Centric Spaces For LiDAR and radar fusion, voxel-centric approaches remain prevalent. * **Sparse Voxel Fusion:** DSERT-RoLL generates initial 3D proposals via LiDAR-4D Radar BEV fusion and projects RGB, thermal, and event features into a unified sparse voxel space using voxel-centric sampling [46]. This allows for adaptive fusion where heterogeneous camera features are aggregated onto 3D proposals via deformable cross-attention. * **Pillar-Based Representations:** Radar data, being inherently sparse, is often aggregated into pillars. CR3DT aggregates radar points into pillars containing velocity features and concatenates them with lifted image features in BEV space, significantly improving detection and tracking without requiring dense LiDAR [47].

2.2.3. Query-Based and Tokenized Representations

Transformers have introduced query-based representations that decouple fusion from fixed grid structures. * **Object Queries:** MSF3DDETR utilizes learnt object queries to sample multi-scale RGB and BEV LiDAR features via cross-attention, enabling set-to-set prediction without non-maximum suppression (NMS) [48]. * **Unified Token Spaces:** MaskFuser tokenizes image and LiDAR features into a joint sequence, applying masked reconstruction to force encoders to preserve critical information in a unified latent space [49]. This approach aligns modalities deeper than simple feature exchange.

2.3. Advanced Fusion Mechanisms

Recent innovations have focused on dynamic, adaptive, and uncertainty-aware fusion mechanisms to handle real-world variability.

2.3.1. Attention-Based and Transformer Fusion

Attention mechanisms allow models to dynamically weigh the importance of different modalities and spatial regions. * **Cross-Modal Attention:** HRFuser extends the HRFormer paradigm to multi-modal settings using Multi-Window Cross-Attention (MWCA) blocks to fuse features at multiple resolutions, preserving high-resolution details across modalities [50]. ZFusion employs Double Deformable Cross Attention (DDCA) to balance sparse radar and dense vision features in an order-invariant manner [51]. * **Spatio-Temporal Attention:** TimeAlign integrates Swin-LSTM to predict current LiDAR BEV features from historical sequences, using camera-guided dual-transformers to correct temporal offsets dynamically [52].

2.3.2. Uncertainty-Aware and Robust Fusion

Quantifying and leveraging uncertainty is crucial for safe fusion, especially under sensor degradation. * **Conformal Prediction:** Cocoon employs learnable surrogate ground truths termed "Feature Impressions" to enable valid conformal prediction, providing uncertainty estimates that adapt to sensor malfunctions like camera blackouts [53]. * **Entropy and Reliability Modeling:** ReliFusion uses Cross-Modality Contrastive Learning to generate modality-specific confidence scores, enabling dynamic fusion weighting that preserves accuracy during sensor degradation [3]. HyperDUM projects latent features into hyperdimensional space to estimate epistemic uncertainty in a single pass, comparable to Bayesian ensembles [54]. * **Mixture of Experts (MoE):** MoME deploys parallel expert decoders for LiDAR-only, camera-only, and fused features, using an Adaptive Query Router to select optimal modalities per query, preventing error propagation from failed sensors [55]. Grace-BEV similarly uses a TrustGate Router to dynamically interpolate between LiDAR-guided and vision-only experts based on geometric integrity [56].

2.3.3. Context-Aware and Adaptive Fusion

Static fusion weights are suboptimal in dynamic environments. Context-aware mechanisms adapt fusion strategies based on environmental conditions. * **Environmental Gating:** ContextualFusion integrates a Gated Convolutional Fusion module that accepts binary context vectors (day/night, rain/clear) to dynamically re-weight sensor modalities, prioritizing LiDAR in low-light conditions [57]. EcoFusion employs a context-aware gating model to predict fusion loss and select Pareto-optimal sensor configurations for energy efficiency [58]. * **Vision-Language Conditioning:** VLC Fusion leverages Vision-Language Models (VLMs) to extract environmental conditions (e.g., "heavy rain," "tunnel") and uses FiLM modulation to dynamically adjust sensor weights, demonstrating that nuanced meta-information improves robustness [59].

2.4. Integration of Novel Sensor Modalities

The sensor suite for autonomous driving is expanding beyond the traditional camera-LiDAR-radar triad to include 4D imaging radar, event cameras, and thermal sensors.

2.4.1. 4D Imaging Radar 4D radar adds elevation and dense Doppler velocity, offering a cost-effective alternative to LiDAR with superior weather robustness. * **Radar-Camera Fusion:** LXL and LXLv2 demonstrate that 4D radar occupancy can assist depth-based sampling for view transformation, achieving performance comparable to LiDAR-based methods in certain scenarios [60], [61]. RCGDet3D introduces Ray-centric Gaussian Splatting to enhance radar feature encoding, achieving real-time perception speeds [62]. * **Cooperative Radar:** In cooperative settings, 4D radar provides weather-resilient Doppler cues that guide feature attention towards moving objects even when LiDAR data is degraded by fog [63].

2.4.2. Event Cameras and Thermal Imaging Event cameras offer microsecond latency and high dynamic range, while thermal sensors provide visibility in total darkness. * **Event-Based Fusion:** InterFuserDVS integrates Dynamic Vision Sensor (DVS) event frames as a third modality, using cross-modal weight initialization to leverage pre-trained RGB models for safe decision-making [64]. DeepIPCv3 utilizes transformer-based cross-modal attention to weight asynchronous DVS events against dense LiDAR geometry for sudden obstacle avoidance [65]. * **Thermal and Multi-Spectral:** DSERT-RoLL incorporates thermal and event cameras alongside stereo RGB and 4D radar, demonstrating that unified voxel-space fusion significantly outperforms standard configurations in fog and snow [46]. SAMFusion utilizes gated NIR cameras to provide superior signal-to-noise ratios in low light compared to passive RGB [66].

2.5. Cooperative and Vehicle-to-Everything (V2X) Perception

Cooperative perception extends the perceptual range beyond line-of-sight by sharing data between vehicles and infrastructure.

2.5.1. Fusion Levels in V2X

- **Intermediate Fusion:** This is the most common approach, balancing bandwidth and accuracy. CoBEVFusion encodes LiDAR and camera data into unified BEV representations locally and fuses them via Dual Window-based Cross-Attention [67]. UniMM-V2X enables cooperation across both perception and prediction levels using Mixture-of-Experts for task-adaptive feature extraction [68].
- **Raw Data Fusion:** While bandwidth-intensive, raw data fusion minimizes information loss. MultiTest and cooperative frameworks demonstrate that merging raw LiDAR point clouds from infrastructure and vehicles significantly reduces occlusion-induced miss detections [69], [70].

2.5.2. Handling Heterogeneity and Misalignment

- **Domain Adaptation:** Heterogeneous sensor setups (e.g., different LiDAR models) introduce domain gaps. S2S-Net addresses this by using sparse voxel grids as a unified representation immune to feature domain gaps, employing scatter operations to merge data from diverse sensors [71].

- **Alignment and Calibration:** AgentAlign constructs a Cross-Modality Feature Alignment Space to mitigate modality-specific noise and misalignment in V2X scenarios [72]. NLOS Dies Twice proposes relay selection policies based on trajectory prediction to maintain connectivity in non-line-of-sight conditions [73].

2.6. Temporal Fusion and Asynchrony Handling

Real-world sensors operate at different frequencies and may suffer from latency, necessitating robust temporal fusion.

2.6.1. Temporal Alignment

- **Explicit Modeling:** TimeAlign explicitly models temporal misalignment as non-affine transformations, using historical LiDAR frames to predict current states and correct offsets [52]. Robust sensor fusion against staleness integrates per-point timestamp offset features to allow models to internally compensate for delays [18].
- **Velocity-Driven Inference:** Velocity Driven Vision utilizes radar velocity data to extrapolate object positions during latency gaps, mitigating asynchrony effects better than static point replication [74].

2.6.2. Spatio-Temporal Fusion

- **Recurrent Architectures:** LEF (Late-to-Early Fusion) recurrently fuses late-stage sparse pillar features from history frames into early stages of the current frame, combining the efficiency of late fusion with the representational power of early fusion [75].
- **Transformer Temporal Modeling:** BEVFusion4D applies ego-motion calibration followed by recurrent deformable attention on concatenated historical frames to achieve motion-aware feature aggregation [43].

2.7. End-to-End Perception and Planning

The trend towards end-to-end learning integrates perception directly with planning and control, bypassing modular hand-offs.

2.7.1. Unified Architectures

- **Perception-Planning Integration:** TransFuser and MaskFuser utilize transformer-based fusion to generate waypoints directly from multi-modal inputs, demonstrating that global contextual reasoning improves collision avoidance [76], [49]. FusionAD constructs a unified BEV multi-modality framework jointly optimizing perception, prediction, and planning, showing significant reductions in displacement error [77].
- **Latent Representation Learning:** DeepSTEP employs sensor-individual encoders fused via self-attention to incorporate temporal dependencies into a shared latent space for joint detection and local mapping [78].

2.7.2. Imitation and Reinforcement Learning

- **Behavior Cloning:** LeTFuser employs a dual-module architecture with Convolutional Vision Transformers for RGB-D fusion, utilizing imitation learning to map sensory inputs to control commands [79].

- **Deep RL:** Hybrid-PPO fuses predicted trajectories and semantic image features into a hybrid state space for Proximal Policy Optimization, mitigating collision risks by anticipating neighbor movements [80].

2.8. Calibration and Sensor Configuration

Accurate calibration is a prerequisite for effective fusion. Recent works focus on online, targetless, and robust calibration methods.

2.8.1. Online and Targetless Calibration

- **Deep Learning Approaches:** RLCNet performs end-to-end deep regression for simultaneous online calibration of LiDAR, radar, and camera, utilizing soft-mask feature sharing to model inter-sensor dependencies [81]. SceneCalib eliminates explicit cross-modal feature correspondence by minimizing reprojection errors of image rays intersecting LiDAR-derived planar surfaces [82].
- **Semantic-Guided Methods:** A Re-Calibration Method uses semantic segmentation consistency across modalities as a supervisory signal to estimate extrinsic parameter biases without requiring explicit targets [83].

2.8.2. Configuration Optimization

- **Information Theoretic Metrics:** Perception Entropy models perception potential via conditional entropy, providing a metric to quantify configuration quality and sensor placement [84]. Unified Surrogate Metrics correlate detection performance with information gain derived from sensor ray coverage, guiding optimal sensor placement [85].
- **Roadside Optimization:** Greedy algorithms based on perceptual gain maximize the placement of roadside LiDARs to minimize blind spots in cooperative scenarios [86].

3. Challenges and Outlook

Despite remarkable advancements, multi-sensor fusion for autonomous driving faces several persistent challenges that hinder widespread deployment. This section discusses key limitations, safety concerns, and future research directions identified in the literature.

3.1. Robustness to Corruption and Adverse Conditions

Sensor performance degrades significantly under adverse weather and corruption, posing a major hurdle for reliability. * **Weather Degradation:** LiDAR suffers from signal attenuation in rain and fog, while cameras fail in low light and glare [2], [4]. Although fusion improves robustness, studies show that dual-source disruptions (e.g., simultaneous LiDAR and camera corruption) cause severe performance drops compared to single-sensor failures [21], [22]. * **Corruption Benchmarks:** Benchmarks like MSC-Bench and MultiCorrupt highlight that current models are fragile to synthetic corruptions such as snow, fog, and sensor misalignment. Asymmetric fusion strategies and modality-independent branches are emerging as necessary defenses [21], [87]. * **Sensor Staleness:** Temporal misalignment due to sensor staleness or latency can lead to catastrophic failures. Methods explicitly encoding time offsets or using velocity-based extrapolation are critical but not yet universally adopted [18], [74].

3.2. Security and Adversarial Vulnerabilities

Multi-sensor fusion systems introduce new attack surfaces, particularly through cross-modal inconsistencies. * **Physical-World Attacks:** MSF-ADV demonstrates that shape manipulation of 3D objects can induce perturbations in both LiDAR and camera inputs simultaneously, bypassing fusion defenses [29]. Frustum attacks exploit the camera's inability to resolve range by injecting spoofed LiDAR points within the camera-defined frustum, maintaining semantic consistency to fool fusion models [88]. * **Temporal Attacks:** DEJAVU injects malicious delays into sensor streams to break temporal synchronization assumptions, causing significant drops in detection and tracking performance [89]. * **Defense Mechanisms:** Neuro-symbolic approaches like NeuSPaPer use scene graph generation to detect semantic structure alterations invisible to standard fusion [90]. However, comprehensive defense strategies against coordinated cross-modal attacks remain an open problem [30].

3.3. Dataset Limitations and Sim-to-Real Gaps

The quality and diversity of datasets directly impact model generalization. * **Data Scarcity:** Many datasets lack comprehensive coverage of adverse weather, night conditions, and rare corner cases [91], [92]. While datasets like MUSES and OmniHD-Scenes integrate event cameras and 4D radar, they are still limited in scale compared to established benchmarks [23], [26]. * **Sim-to-Real Gap:** Models trained on simulated data (e.g., CARLA) often fail to generalize to real-world noise and sensor characteristics [93], [94]. Collaborative perception datasets like Mixed Signals highlight significant domain gaps when transferring models trained on right-hand traffic to left-hand scenarios or heterogeneous sensor setups [95]. * **Annotation Challenges:** Generating ground truth for novel sensors like 4D radar and event cameras is difficult. Automated labeling frameworks like RALF and LiCROcc use cross-modal distillation to generate pseudo-labels, but errors persist in complex scenarios [96], [97].

3.4. Computational Efficiency and Deployment

Real-time constraints on embedded hardware limit the complexity of fusion architectures. * **Latency:** Transformer-based fusion and diffusion models often incur high computational costs, conflicting with real-time requirements [98], [99]. Lightweight architectures like NextBEV and depthwise optimizations are necessary to reduce latency without sacrificing accuracy [100]. * **Energy Efficiency:** EcoFusion and similar adaptive methods highlight the need for energy-aware fusion, dynamically selecting sensor subsets based on context to reduce power consumption [58]. * **Edge Computing:** In V2X scenarios, bandwidth constraints necessitate efficient compression. Autoencoders and sparse evidence retrieval (e.g., INTACT) offer promising directions for reducing communication overhead while maintaining perception quality [101], [102].

3.5. Future Directions

- **Foundation Models and LLMs:** Integrating Vision-Language Models (VLMs) and Large Language Models (LLMs) for context-aware reasoning and zero-shot adaptation is an emerging frontier. PLA frameworks convert sensor data into structured text for LLM-based planning, though latency and hallucination risks remain concerns [98], [103].
- **Occupancy Networks:** Moving beyond bounding boxes to 3D semantic occupancy prediction provides a denser, more complete scene representation. Methods like OccFusion and DAOcc demonstrate the benefits of multi-modal fusion for occupancy tasks, especially in

dynamic and occluded scenarios [104], [105].

- **Neuro-Symbolic Integration:** Combining deep learning with symbolic reasoning and physics-based constraints can enhance interpretability and safety, particularly for detecting stealthy attacks and corner cases [90], [106].
- **Standardization and Safety Compliance:** Developing standardized evaluation metrics for robustness, safety, and uncertainty is crucial. Frameworks aligning with ISO 26262, such as safety-compliant transformer architectures, are needed to ensure functional safety in deployment [106].

4. Conclusion

Multi-sensor fusion has established itself as an indispensable component of autonomous driving perception, enabling systems to overcome the limitations of individual sensors through complementary redundancy. This survey has synthesized a vast body of literature, highlighting the evolution from simple late fusion to sophisticated transformer-based, BEV-centric, and uncertainty-aware architectures. The integration of novel modalities like 4D radar and event cameras, coupled with the rise of cooperative V2X perception, promises to further enhance robustness and range.

However, significant challenges remain in ensuring robustness against adverse weather, adversarial attacks, and sensor failures. The field must address the sim-to-real gap, improve computational efficiency for embedded deployment, and develop rigorous safety standards. Future research should focus on foundation models for generalization, neuro-symbolic approaches for interpretability, and comprehensive benchmarking under realistic corruption scenarios. As autonomous vehicles move towards widespread deployment, the development of resilient, adaptive, and safe multi-sensor fusion systems will be paramount to achieving the vision of fully autonomous mobility.

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